



Identification of sensor block model using Volterra series and frequency response function

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ABSTRACT

This paper proposes a kind of identification method of block models to describe the nonlinear dynamic characteristics of sensors. The Volterra series are employed to analyze the block models and separate the higher-order Volterra kernels of the block models. The estimation method of frequency response function in the non-parametric form is used to identify the model of linear dynamic subsystem. And then coefficients of the nonlinear static subsystem are determined according to the nonlinear outputs of various orders. The identification method only needs the step or impact signal with different amplitudes as the input. These experimental data can be obtained in the calibration experiments of sensors easily. Considering the random noise or Gaussian noise in calibration experiments, the correlation function and bispectrum are used to reduce the influence of noise and improve the calculation precision. The method is applied to the hot-film mass air flow sensor for building its nonlinear dynamic model.

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1. Introduction

There are four kinds of methods to describe the nonlinear dynamic characteristics of sensors. (1) System identification method. Although it is applied successfully to the linear system, this method is very difficult to identify the parameters of nonlinear system because there are many higher-order items and cross items in the difference equation [1]. (2) Functional series such as the Volterra series and Wiener series. These series can be understood easily and show obvious physical concepts, but are difficult to be identified because it is very hard to extract higher-order kernels from their integer expressions. Usually this problem is resolved through constructing random signals with special properties as the input, such as Gaussian signal or the pseudorandom signal. Based on these input–output data the nonlinear dynamic model can be identified by the correlation technique, the orthogonal demodulation,

the frequency domain analysis and so on [2–14]. In practice, however, it is difficult to generate a Gaussian signal as the sensor input. In calibrating experiments of sensors, the impact, step and sine signals are often used as the inputs. (3) Artificial neural networks. Since the nonlinear dynamic system is not satisfied with the homogeneity and superposition, its models are different when the forms and amplitudes of input are different. (4) Block models. It consists of a linear dynamic subsystem and a nonlinear static subsystem, and is easy to be identified relatively. This kind of model is suitable for the dynamic nonlinear correction of sensors [15], so this paper studies the identification method of the block model.

The Wiener model and the Hammerstein model are two basic types of block model. The Wiener model is given by the cascade connection of a linear dynamic block followed by a nonlinear static subsystem shown in Fig. 1; the Hammerstein model is given by the cascade connection of a nonlinear static block followed by a linear dynamic subsystem shown in Fig. 2. In these figures, $h(t)$ is the transfer function of the linear dynamic subsystem. The mathematical model of the linear dynamic subsystem is

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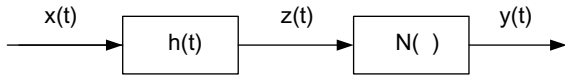


Fig. 1. Block diagram of the Wiener model.

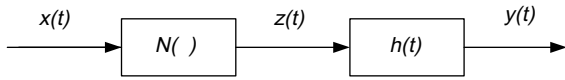


Fig. 2. Block diagram of the Hammerstein model.

$$y(n) = \sum_{i=0}^l b_i x(n-i) - \sum_{j=1}^m a_j y(n-j) \quad (1)$$

where $b = \{b_0, b_1, b_2, \dots, b_l\}$ and $a = \{a_1, a_2, \dots, a_m\}$ are parameters of the linear subsystem, l and m are orders of the subsystem. $N(\cdot) = z(t) + r_2 z^2(t) + r_3 z^3(t) + \dots$ is the approximate expression of the nonlinear static subsystem. Therefore the main task of modeling focuses on identifying parameters of the linear dynamic part and nonlinear static part $r = \{r_1, r_2, \dots, r_n\}$. Voros utilized the least squares method and the artificial neural network to identify parameters of the Hammerstein model [9]. Hu adopted three-level-pseudorandom-sequences with different amplitudes as inputs, and identified the impact response function of the linear part and the polynomial coefficients of the nonlinear part of the discrete-time Wiener model of nonlinear systems [16]. Kozek used the extended Kalman filter (EKF) for parameter identification of Hammerstein/Wiener nonlinear systems [17]. Katoh proposed a Hammerstein model identification method using genetic programming [18]. Bai discussed discrete Hammerstein model identification using a blind system identification approach [19]. Li applied genetic algorithms to the identification of the Hammerstein model. He used piecewise linear method to approximate the memoryless nonlinear characterization [20].

This paper presents a new kind of identification method of block model. We apply the Volterra series to analyzing these models, and find that the higher-order Volterra kernels of block model have parameter separable property. We separate the linear block from the nonlinear block with the multilevel inputs and their corresponding outputs, and then use the linear transfer function in non-parameter form to build the linear dynamic block and a polynomial to approximate the nonlinear static block.

2. Algorithm derivation

2.1. Identification of the Wiener model

In Fig. 1, the relationship between the input and output of the Wiener model can be written as follows:

$$z(t) = \int_{-\infty}^{+\infty} h(\tau) x(t-\tau) d\tau \quad (2)$$

$$y(t) = N[z(t)] = z(t) + r_2 z^2(t) + r_3 z^3(t) + \dots \quad (3)$$

Substituting Eq. (2) into Eq. (3), we can obtain:

$$y(t) = \int_{-\infty}^{+\infty} h(\tau) x(t-\tau) d\tau + r_2 \left[\int_{-\infty}^{+\infty} h(\tau) x(t-\tau) d\tau \right]^2 + r_3 \left[\int_{-\infty}^{+\infty} h(\tau) x(t-\tau) d\tau \right]^3 + \dots \quad (4)$$

This type of the Wiener model can also be described by the Volterra series:

$$y(t) = \int_{-\infty}^{+\infty} h_1(\tau) x(t-\tau) d\tau + \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h_2(\tau_1, \tau_2) x(t-\tau_1) x(t-\tau_2) d\tau_1 d\tau_2 + \dots = \sum_{n=1}^{+\infty} y_n \quad (5)$$

Here, define y_n as the n th order output of the system, for example, y_1 is the first order output and is the linear output of system too, y_2 is the second order output and is a part of nonlinear output of system.

$$y_n = \int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} h_n(\tau_1, \dots, \tau_n) x(t-\tau_1) \dots x(t-\tau_n) d\tau_1 \dots d\tau_n \quad (5a)$$

$h_1(t)$ and $h_2(t_1, t_2)$ are the first order and the second order Volterra kernels, respectively, and $h_n(t_1, \dots, t_n)$ is the n th order Volterra kernel.

Eq. (4) becomes

$$\begin{aligned} y(t) &= \int_{-\infty}^{+\infty} h(\tau) x(t-\tau) d\tau + r_2 \left[\int_{-\infty}^{+\infty} h(\tau_1) x(t-\tau_1) d\tau_1 \right] \\ &\quad \cdot \left[\int_{-\infty}^{+\infty} h(\tau_2) x(t-\tau_2) d\tau_2 \right] \\ &\quad + r_3 \left[\int_{-\infty}^{+\infty} h(\tau_1) x(t-\tau_1) d\tau_1 \right] \\ &\quad \cdot \left[\int_{-\infty}^{+\infty} h(\tau_2) x(t-\tau_2) d\tau_2 \right] \\ &\quad \cdot \left[\int_{-\infty}^{+\infty} h(\tau_3) x(t-\tau_3) d\tau_3 \right] + \dots \\ &= \int_{-\infty}^{+\infty} h(\tau) x(t-\tau) d\tau + \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} r_2 h(\tau_1) h(\tau_2) x \\ &\quad \times (t-\tau_1) x(t-\tau_2) d\tau_1 d\tau_2 \\ &\quad + \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} r_3 h(\tau_1) h(\tau_2) h(\tau_3) x(t-\tau_1) x(t-\tau_2) x \\ &\quad \times (t-\tau_3) d\tau_1 d\tau_2 d\tau_3 + \dots \end{aligned} \quad (6)$$

Comparing Eq. (5) with Eq. (6), we can obtain

$$h_1(t) = h(t) \quad (7a)$$

$$h_2(t_1, t_2) = r_2 h(t_1) h(t_2) \quad (7b)$$

$$h_n(t_1, \dots, t_n) = r_n h(t_1) \dots h(t_n) \quad (7c)$$

Eqs. (7a)–(7c) indicate that the higher-order Volterra kernels are of parameter separable. Considering Eqs. (4) and (6), we have

$$y_n(t) = \int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} r_n h(\tau_1) \dots h(\tau_n) x(t-\tau_1) \dots x(t-\tau_n) d\tau_1 \dots d\tau_n = r_n [y_1(t)]^n \quad (6a)$$

Thus the higher-order outputs of the Wiener model can be computed by the convolution of the higher-order kernel with sensor input, and also be obtained with the first order

output y_1 and r_n . Obviously the calculation of the latter is simpler than that of the former, and higher is the order, more obvious is the advantage.

Define $\{k_1x_0, k_2x_0, \dots, k_nx_0\}$ as a set of multilevel signals, and x_0 is its base signal. According to the relationship between the input and output of the Wiener model, when a set of multilevel signals are inputted to the system, we have

$$\begin{aligned} \text{When } x(t) &= x_0(t), y(t) = y_1(t) + y_2(t) + \dots \\ \text{When } x(t) &= k_1x_0(t), \hat{y}^{(1)}(t) = k_1 \cdot y_1(t) + k_1^2 \cdot y_2(t) + \dots \\ \text{When } x(t) &= k_2x_0(t), \hat{y}^{(2)}(t) = k_2 \cdot y_1(t) + k_2^2 \cdot y_2(t) + \dots \end{aligned}$$

Using $X = \{k_1x_0, k_2x_0, \dots, k_nx_0\}$ to describe the multilevel input signals, the matrix expressing the relationship between the input and output will be

$$\hat{Y} = AY \quad (8)$$

where $\hat{Y} = [\hat{y}^{(1)}, \dots, \hat{y}^{(n)}]^T$, $\hat{y}^{(1)}, \hat{y}^{(2)}, \dots, \hat{y}^{(n)}$ are measured outputs corresponding to inputs $k_1x_0, k_2x_0, \dots, k_nx_0$; $Y = [y_1, y_2, \dots, y_n]^T$ is the vector made up of the i th order output corresponding to the input x_0 ($i = 1, 2, \dots, n$);

$$A = \begin{bmatrix} k_1 & k_1^2 & \dots & k_1^n \\ k_2 & k_2^2 & \dots & k_2^n \\ \dots & \dots & \dots & \dots \\ k_n & k_n^2 & \dots & k_n^n \end{bmatrix}$$

When $k_i \neq \gamma k_j$ ($i, j = 1, 2, \dots, n$; $i \neq j$; $\gamma \in R$) and is not zero, matrix A will be non-singular, then Eq. (8) has only one solution:

$$Y = A^{-1}\hat{Y} \quad (8a)$$

Therefore, if the nonlinear system is excited by a set of multilevel signals, the i th order output $y_i(t)$ ($i = 1, 2, \dots, n$) can be extracted. According to $y_n(t)$ the linear transfer function $h(t)$ and coefficients of the nonlinear static part r_i ($i = 2, 3, \dots$) can be identified.

With $y_1(t)$ being isolated, we will have

$$H(f) = \frac{Y_1(f)}{X_0(f)} \quad (9)$$

where $Y_1(f)$, $X_0(f)$ and $H(f)$ are the Fourier transform of $y_1(t)$, $x_0(t)$ and $h(t)$, respectively, and $h(t)$ will be obtained by the inverse Fourier transform. We have

$$h_{is}(t_1, \dots, t_i) = h(t_1) \dots h(t_i) \quad (9a)$$

$$y_{is}(t) = \int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} h_{is}(\tau_1, \dots, \tau_i) x(t - \tau_1) \dots x(t - \tau_i) d\tau_1 \dots d\tau_i = [y_1]^i \quad (9b)$$

where $h_{is}(t_1, \dots, t_i)$ and y_{is} are reference values of the i th order Volterra kernel $h_i(t_1, \dots, t_i)$ and the i th order output y_i , respectively, which are multiple of $h_i(t_1, \dots, t_n)$ and y_i . Comparing with (6a) and (7c), there will be:

$$\begin{aligned} h_i(t_1, \dots, t_i) &= r_i h_{is}(t_1, \dots, t_i) \\ y_i(t) &= r_i y_{is}(t) \end{aligned} \quad (10)$$

$$r_i = \frac{y_i}{y_{is}} \quad (i = 1, 2, \dots, n)$$

Therefore the identifying procedure of identifying the Wiener model is presented as follows:

- (1) A set of multilevel signals $X = [k_1x_0, k_2x_0, \dots, k_nx_0]^T$ are inputted to the system, and outputs $\hat{Y} = [\hat{y}^{(1)}, \dots, \hat{y}^{(n)}]^T$ are obtained. Then y_i ($i = 1, 2, \dots, n$) are extracted by Eq. (8a);
- (2) With $y_1(t)$ and $x_0(t)$, the linear transfer function $h(t)$ and y_{is} are obtained according to Eqs. (9) and (9b);
- (3) Eq. (10) is used to calculate r_i ($i = 1, 2, \dots, n$).

2.2. Identification of the Hammerstein model

A block diagram of the Hammerstein model is shown in Fig. 2. The relationship between the input and output can be described as follows:

$$\begin{aligned} z(t) &= x(t) + r_2x^2(t) + r_3x^3(t) + \dots \\ y(t) &= \int_{-\infty}^{+\infty} h(\tau)z(t - \tau) d\tau \\ &= \int_{-\infty}^{+\infty} h(\tau)x(t - \tau) d\tau + r_2 \int_{-\infty}^{+\infty} h(\tau)x^2(t - \tau) d\tau \\ &\quad + r_3 \int_{-\infty}^{+\infty} h(\tau)x^3(t - \tau) d\tau + \dots \\ &= \sum_{i=1}^{+\infty} y_i \end{aligned}$$

where $y_i = r_i \int_{-\infty}^{+\infty} h(\tau)x^i(t - \tau) d\tau$ ($r_1 = 1$), y_i is the i th order output of the Hammerstein model. Similar to the Wiener model, when the system based on the Hammerstein model is excited by a set of multilevel signals $X = [k_1x_0, k_2x_0, \dots, k_nx_0]^T$, the measured output are $\hat{Y} = [\hat{y}^{(1)}, \hat{y}^{(2)}, \dots, \hat{y}^{(n)}]^T$. The output of various orders y_i ($i = 1, 2, \dots, n$) can be extracted according to Eq. (8a).

With y_1 and x_0 , $h(t)$ can be calculated through Eq. (9). Then,

$$y_{is}(t) = \int_{-\infty}^{+\infty} h(\tau)x_0^i(t - \tau) d\tau$$

where y_{is} is the reference value of the i th order output of the Hammerstein model.

According to Eq. (10), r_i ($i = 2, 3, \dots, n$) can be obtained. The identifying process of the Hammerstein model is the same as that of the Wiener model.

3. Algorithm improvement

In practice there always is noise in the sensor output. If an additive noise is added to the output of the Wiener model and the Hammerstein model, their block diagrams will be described as in Figs. 3 and 4. Different from Figs. 1 and 2, the outputs in Figs. 3 and 4 contain the noise $n(t)$ besides the sensor output. Since the identifying procedure and the equations of the Hammerstein model are similar to those of the Wiener model, we only discuss the Wiener model in the following part. In the following equation, $\xi(t)$ is used to describe the noise taking the place of $n(t)$:

$$y(t) = \sum_{i=1}^{+\infty} y_i + \xi(t) \quad (11)$$

Eq. (8) can also be:

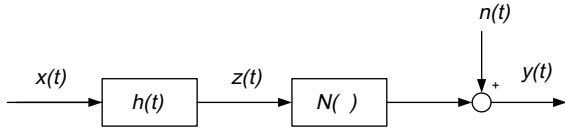


Fig. 3. Block diagram of the Wiener model (noise).

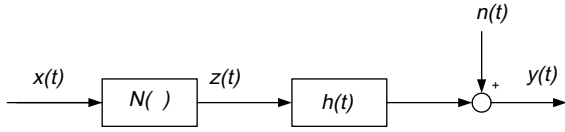


Fig. 4. Block diagram of the Hammerstein model (noise).

$$\hat{Y} = AY + \xi \quad (12)$$

The vector $\hat{Y} = [\hat{y}^{(1)}, \dots, \hat{y}^{(n)}]^T$ is the corresponding to the input $X = [k_1x_0, k_2x_0, \dots, k_nx_0]^T$, and y_i ($i = 1, 2, \dots, n$) of the vector $Y = [y_1, y_2, \dots, y_n]^T$ is the i th order output of the input x_0 , ξ_i ($i = 1, 2, \dots, n$) of the vector $\xi = [\xi_1, \xi_2, \dots, \xi_n]^T$ is the noise of the i th experiment. Since the noise can affect results of extracting the various degree outputs from nonlinear system outputs, the correlation function and the bispectrum are applied to improve the identification method mentioned above, aiming at reducing the noise influence, only considering random noise and the Gaussian noise.

3.1. Identification method improved by the correlation technique

The correlation technique can extract out the useful signal from the mixed signal including noises. Here the improved method performs the correlation of the input with output, and can reduce the noise disturbance because of the independence of signal and noise.

The correlation between the input and output of Eq. (11) is

$$\begin{aligned} R_{xy}(m) &= \frac{1}{T} \int_{t=-\infty}^{+\infty} x(t)y(t+m) dt \\ &= \sum_{i=1}^{+\infty} R_{xy_i}(m) + R_{x\xi}(m) \end{aligned} \quad (13)$$

The input $x(t)$ and the noise $\xi(t)$ are independent, then

$$\begin{aligned} R_{x\xi}(m) &= 0 \\ R_{xy}(m) &= \sum_{i=1}^{+\infty} R_{xy_i}(m) \end{aligned}$$

The correlation between the reference input $x_0(t)$ and two sides of Eq. (12) is

$$R_{x_0\hat{Y}} = AR_{x_0Y} \quad (13a)$$

where $R_{x_0\hat{Y}} = [R_{x_0\hat{y}^{(1)}}, \dots, R_{x_0\hat{y}^{(n)}}]^T$; $R_{x_0Y} = [R_{x_0y_1}, \dots, R_{x_0y_n}]^T$; $R_{x_0\hat{y}^{(i)}}$ is the correlation of the input x_0 with the i th output $\hat{y}^{(i)}$ corresponding to the input k_ix_0 , and $R_{x_0y_i}$ is the correlation of the input x_0 with its i th order output y_i .

Thus Eqs. (9) and (10) become

$$H(f) = \frac{P_{x_0y_1}}{P_{x_0}} \quad (14)$$

$$r_i = \frac{R_{x_0y_i}}{R_{x_0y_{is}}} \quad (15)$$

where $P_{x_0y_1}$ is the cross spectrum between the input x_0 and the first order output (linear output) y_1 , and is the Fourier transform of the cross correlation $R_{x_0y_1}$ too; P_{x_0} is the auto spectrum of the input x_0 ; $R_{x_0y_{is}}$ is the cross correlation of the input x_0 with the i th order reference output y_{is} , which can be computed by Eq. (9b).

Therefore the general procedure of identifying the Wiener model with the improved method by the autocorrelation technique can be described as follows:

- (1) A suitable base input $x_0(t)$ is chosen. A set of multi-level signals $X = [k_1x_0, k_2x_0, \dots, k_nx_0]^T$ are inputted to the nonlinear system, and outputs are $\hat{Y} = [\hat{y}^{(1)}, \hat{y}^{(2)}, \dots, \hat{y}^{(n)}]^T$. The input–output data is pre-treated, then $R_{x_0\hat{Y}} = [R_{x_0\hat{y}^{(1)}}, R_{x_0\hat{y}^{(2)}}, \dots, R_{x_0\hat{y}^{(n)}}]^T$ is calculated, and $R_{x_0Y} = [R_{x_0y_1}, R_{x_0y_2}, \dots, R_{x_0y_n}]^T$ is computed by Eq. (13a);
- (2) The transfer function $h(t)$ of linear part is calculated through Eq. (14), and the y_{is} ($i = 2, 3, \dots, n$) are computed by Eq. (9a);
- (3) The cross correlation $R_{x_0y_{is}}$ of x_0 with y_{is} is calculated, and r_i ($i = 2, 3, \dots, n$) are determined by Eq. (15).

3.2. Identification method improved by bispectrum

Since most noises are Gaussian distributed in practical cases, they can be restrained by the higher-order spectrum, the identification method of the block model can be improved by the higher-order spectrum. The bispectrum is the Fourier transform of three-order correlation, simpler to be computed, and applied wider, which can be defined as follows:

$$\begin{aligned} B_x(f_1, f_2) &= X^*(f_1 + f_2)X(f_1)X(f_2) \\ B_{xy}(f_1, f_2) &= X^*(f_1 + f_2)X(f_1)Y(f_2) \\ B_x(f, f) &= X^*(2f)X(f)X(f) \\ B_{xy}(f, f) &= X^*(2f)X(f)Y(f) \end{aligned}$$

where $B_x(f_1, f_2)$ is the auto-bispectrum of x , and $B_x(f, f)$ is its diagonal slide; $B_{xy}(f_1, f_2)$ is the cross bispectrum between x and y , and $B_{xy}(f, f)$ is its diagonal slide.

So the frequency domain form of Eq. (11) becomes

$$Y(f) = \sum_{i=1}^{+\infty} Y_i(f) + \bar{\xi}(f) \quad (16)$$

where $Y(f)$, $Y_i(f)$ and $\bar{\xi}(f)$ are the Fourier transform of $y(t)$, $y_i(t)$ and $\xi(t)$, respectively. The frequency domain form of Eq. (12) becomes

$$\hat{Y}_f = AY_f + \bar{\xi} \quad (17)$$

where $\hat{Y}_f$, Y_f and $\bar{\xi}$ are the Fourier transform of $\hat{Y}$, Y and ξ . Taking the cross bispectrum between the input and output for two sides of Eq. (16), we have

$$B_{xy}(f, f) = \sum_{i=1}^{+\infty} B_{xy_i}(f, f) + B_{x\xi}(f, f) \quad (16a)$$

If $\xi(t)$ is a Gaussian noise, and independent to the input $x(t)$, so $B_{xx\xi}(f,f) = 0$,

$$B_{xxy}(f,f) = \sum_{i=1}^{+\infty} B_{xxy_i}(f,f) \quad (16c)$$

If the input $x(t)$ has some Gaussian noises independent to it, Eq. (17) still works.

Taking the cross bispectrum for Eq. (17), we obtain

$$B_{x_0x_0y}(f,f) = AB_{x_0x_0y}(f,f) \quad (17a)$$

where $B_{x_0x_0y} = [B_{x_0x_0y^{(1)}}, B_{x_0x_0y^{(2)}}, \dots, B_{x_0x_0y^{(n)}}]^T$, and $B_{x_0x_0y} = [B_{x_0x_0y_1}, B_{x_0x_0y_2}, \dots, B_{x_0x_0y_n}]^T$.

Therefore if the system is excited by a set of multilevel signals $X = [k_1x_0, k_2x_0, \dots, k_nx_0]^T$ and $\hat{Y} = [\hat{y}^{(1)}, \hat{y}^{(2)}, \dots, \hat{y}^{(n)}]^T$ are their outputs. The cross bispectrum between the input and output is calculated. $B_{x_0x_0y} = [B_{x_0x_0y_1}, B_{x_0x_0y_2}, \dots, B_{x_0x_0y_n}]^T$ can be extracted by Eq. (17a), where $y_i (i=1, 2, \dots, n)$ is the i th order output corresponding to the input x_0 . So we can have

$$H(f) = \frac{B_{x_0x_0y_1}(f,f)}{B_{x_0}(f,f)} \quad (18)$$

$$r_i = \frac{B_{x_0x_0y_i}}{B_{x_0x_0y_n}} \quad (19)$$

The identifying procedure of the Wiener model by the improved method with bispectrum is the similar as those of autocorrelation's in which the only requirement is that the bispectrum takes the place of the autocorrelation.

4. Simulation results

The simulations are carried out to verify the effectiveness of the algorithms. A linear part $h(t)$ and a nonlinear part $N(\cdot)$ are constructed, and with their combination in different sequence the Wiener model and the Hammerstein model can be gained. Parameters of the linear part are selected as: $b = [0.0010 \ 0.1119 \ 0.0010]$, $a = [1.0000 \ -1.9874 \ 0.9913]$; for the nonlinear static part we choose two subsection expressions of the positive direction and the negative direction:

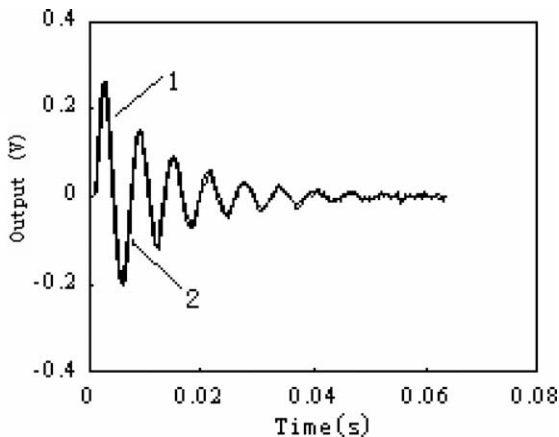


Fig. 5. Impact response of Wiener model 1 the real output, 2 the modeling output (improved by correlation).

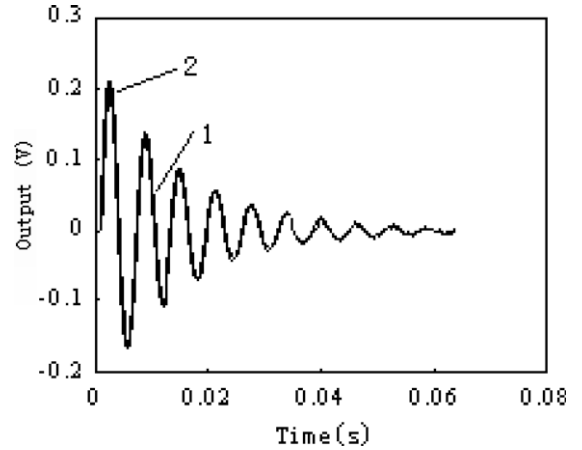


Fig. 6. Impact response of Hammerstein model 1 the real output, 2 the modeling output (improved by correlation).

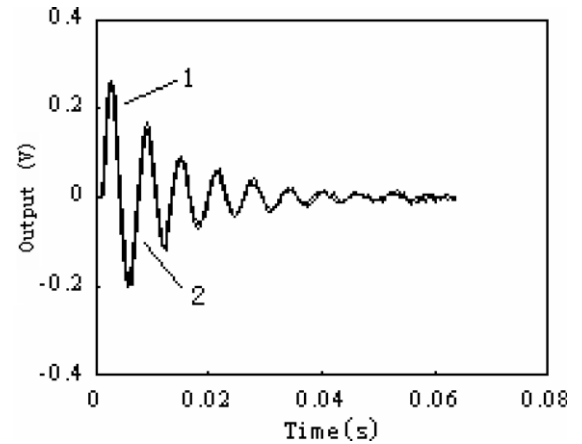


Fig. 7. Unit impact response of Wiener model 1 the real output, 2 the modeling output (improved by bispectrum).

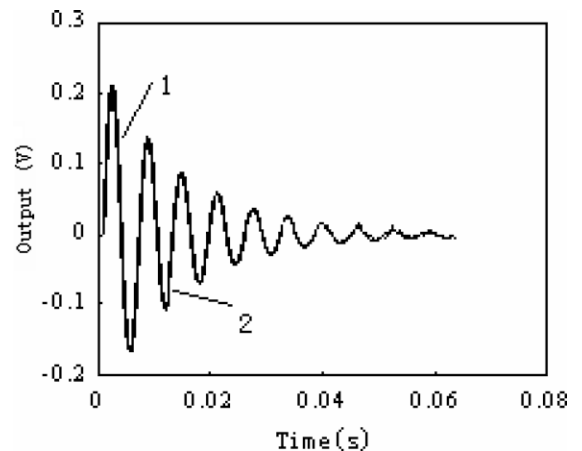


Fig. 8. Unit impact response of Hammerstein model 1 the real output, 2 the modeling output (improved by bispectrum).

$$N[z(t)] = z(t) + r_2 z^2(t), \quad \text{when } z(t) \geq 0, \\ r_2 = 2, \quad \text{when } z(t) < 0, \quad r_2 = -2.$$

When the noise is added to the sensor output, results of modeling are shown in Figs. 5–8. In the simulation the unit impact signal is adopted as the fundamental signal of the required multilevel inputs. The noise is generated by functions of MATLAB. From these figures the model output is identical with the real output, and it also can be noticed that there exist fluctuations in the model output, which is because the cross correlation function and bispectrum cannot completely get rid of noise contained in the output, but the results in above figures can meet the requirements.

5. Experimental results

The hot-film mass air flow (MAF) sensors based on a thermal heat-loss principle are used in the automotive engines to measure the intake mass air flow exactly and control the air-to-fuel ratio precisely. Since the speed of air entering the engines with four or fewer cylinders is very fast, the hot-film MAF sensor should possess good dynamic performance. The results of static calibration show the hot-film/wire MAF sensors are the nonlinear devices [21]. Furthermore the dynamic experiments indicate that there is also the dynamic non-linearity in their responses, which affects their measurement accuracy. Hence it is important to study the dynamic non-linearity of hot-film MAF sensors. First of all, the dynamic calibration experiments were performed, and the step input and output with different amplitudes were collected. And then the nonlinear dynamic models were built by the method proposed by this paper.

5.1. Experiments

5.1.1. Static calibration experiments

The experimental facility shown in Fig. 9 was used to examine the MAF sensor static characteristic. The experimental setup consists of an air pump with 15 kW, an air tank with 2 m³, a straight pipe with the diameter 60 mm,

a regulating valve, a HFM5 type of the hot-film MAF sensor made by Bosch Investment Ltd., a CLC-100 type laminar flowmeter with an YJB-1500 type micro-manometer, and test equipments. The test equipments include: a MS8050 digital voltmeter and a TDS2024 digital scope to measure and log the sensor output; a digital thermometer to test the intake air temperature at the in port of the pipe; a free air thermometer, an atmospheric pressure meter and a hygrometer to record the environmental parameters.

The air stream passed through the laminar flowmeter, MAF sensor and the valve, and then entered into the air tank. Finally, the air stream was drawn out of the air tank by the air pump. The air flow-rate was measured by the laminar flowmeter, and displayed with the micro-manometer in the form of the different pressure whose unit was mm water-column. During the static calibration experiments, 35 testing points were selected within the measuring range from 0 to 91.5778 g/s. When the readings of micro-manometer were between 0 and 10 mm water-column, a point was chosen every other 1 mm water-column. Between 12 and 20 mm, a point was chosen every other 2 mm. Between 25 and 120 mm, a point was chosen every other 5 mm. When the experiments began, the zero point of the micro-manometer was adjusted, and the zero point of the MAF sensor was recorded. Then the air pump was turned on, the air flow-rate was adjusted by the flow-rate regulating valve so as to reach a series of fixed points of the micro-manometer. At the same time, all parameters were recorded manually, such as the different pressure, atmospheric pressure, environment temperature, atmospheric humidity and intake air stream temperature. The corresponding air flow-rate was calculated according to all these parameters. After the positive travel of static experiment (the flow-rate run from small to large) was completed, the negative travel (the flow-rate was from large to small) was conducted, and the flow-rate points were selected as same as those of the positive travel.

5.1.2. Dynamic calibration experiments

Fig. 10 is a block diagram of the dynamic experimental setup. The fast manual valve was a quick conical triple

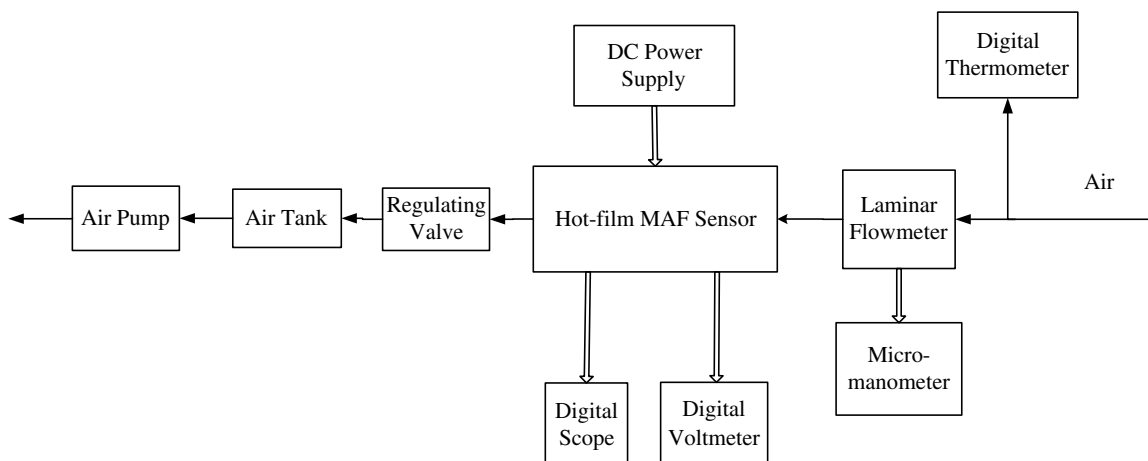


Fig. 9. Block diagram of static calibration experimental facility.

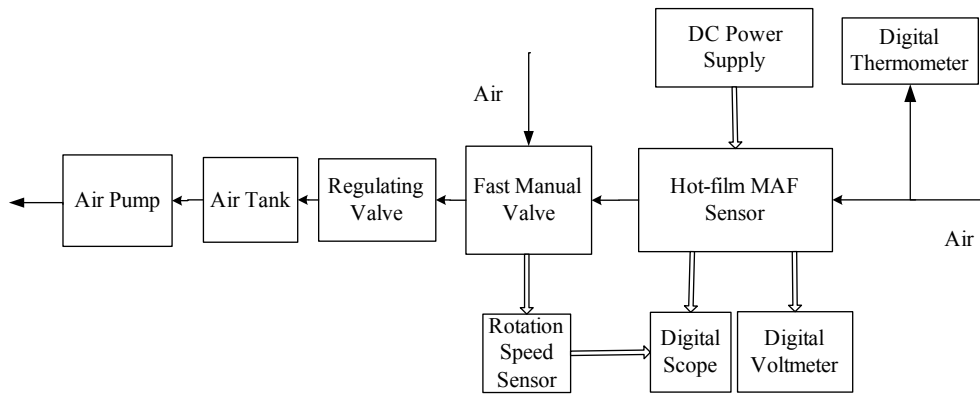


Fig. 10. Block diagram of dynamic experimental setup.

valve made by ourselves, and could turn off/on the air stream rapidly to generate the step signal. When it was turned off, its one port is opened to the atmosphere, and the other port connecting to the pipe was closed. The air stream did not pass through the MAF sensor. When the fast manual valve was turned on, the air stream passed through the MAF sensor, fast manual valve and flow-rate regulating valve, and then entered in the air tank. An electromagnetic type rotate speed sensor was installed in the fast manual valve, and detected the action of the valve. When the fast manual valve was switch on/off, the rotate speed sensor generated an impulse signal simultaneously to indicate the starting point of dynamic response of the MAF sensor. Two channels of the digital scope collected the signals of both the MAF sensor and rotate speed sensor at the same time.

Some flow-rate points were chosen to perform dynamic calibration experiments. Since the static calibration experiments of these points were conducted, the corresponding relationship between the input flow-rate and MAF sensor output has been known. We could adjust the input flow-rate according the MAF sensor output without the laminar flowmeter. The experimental procedure was as follows. (1) Adjusting the regulating valve to make the flow-rate reach the setting value. (2) Pressing the button of fast manual valve to close the valve rapidly, which generates a negative step flow-rate signal. (3) Both the MAF sensor response and rotation speed sensor output were collected by the digital scope, and sent into a personal computer. Thus, a negative step experiment was completed. (4) The handle of the fast manual valve was turned to the other direction. The button of the fast manual valve was pressed again, and the valve was open rapidly. A positive step flow-rate signal was formed. (5) The experimental data were logged.

5.2. Modeling results

According to the experimental data the nonlinear dynamic model of the MAF sensor was built by means of the method proposed by this paper. The comparison between the model outputs and the actual responses with

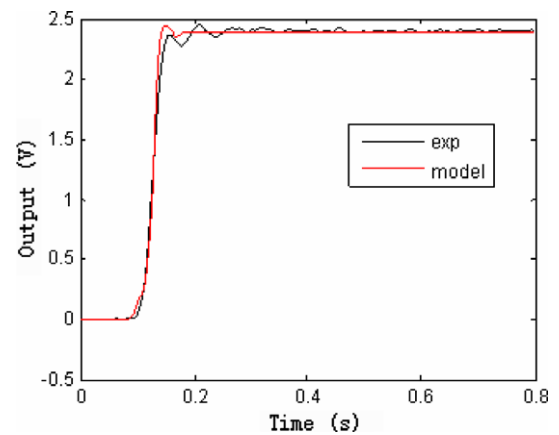


Fig. 11. Comparison between actual response and model output under input 37.9719 g/s.

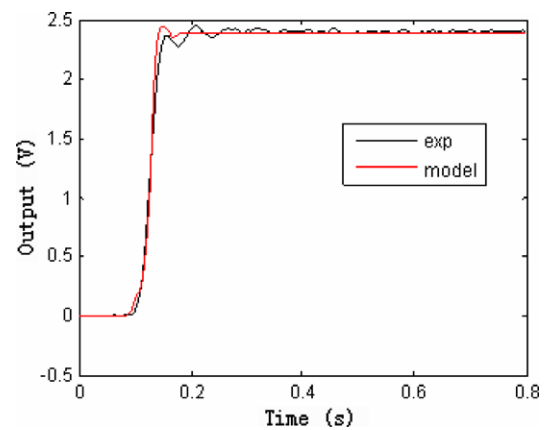


Fig. 12. Comparison between actual response and model output under input 66.8078 g/s.

different amplitudes are shown in Figs. 11–13 in order to verify the validity of the method.

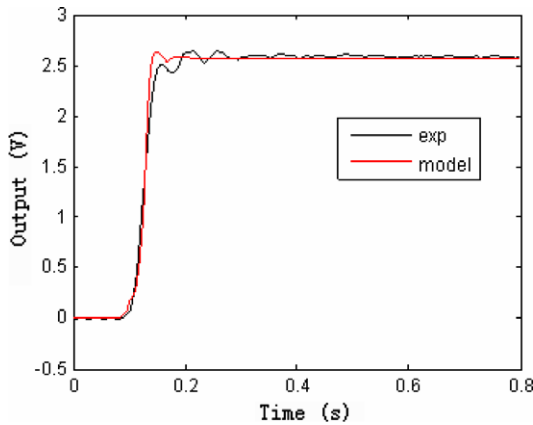


Fig. 13. Comparison between actual response and model output under input 84.5464 g/s.

6. Conclusions

- (1) The order of the nonlinear static part will affect the modeling precision, and theoretically higher the order is, more accurate the model is. Therefore a suitable order should be prefixed under the assumption that error to some extent is permitted. In addition, the most nonlinear systems should be established in two directions: the negative and the positive, which had also been analyzed in [9]. In this paper, the simulation modeling in two directions for nonlinear part is performed.
- (2) The method requires a set of multilevel signals as inputs to extract various order outputs corresponding to the input x_0 , and the number of multilevel signals should be equal to the order of the nonlinear static part. At the same time matrix A should be ensured to be non-singular, then Eq. (8) has one solution.
- (3) The linear transfer function should be identified more accurately because it will also affect the estimation of the nonlinear part parameters. The linear output extracting from total output is

$$\bar{y}_1 = y_1 + \zeta_1$$

where $\bar{y}_1$ is an estimate value of the linear output, y_1 is the real linear output, ζ_1 is the noise, and the identification of $h(t)$ will be greatly influenced by noises. The identification formula of nonlinear part parameters is

$$r_i = \frac{y_i}{y_{is}}$$

where y_i and y_{is} are the i th order output and i th order reference output, in fact the average formula is adopted in order to reduce the affection of noises and avoid the possibility of zero divisor:

$$r_i = \frac{\text{mean}(y_i)}{\text{mean}(y_{is})}$$

where $\text{mean}(\cdot)$ is an average value of vector. If selecting one part of vector in which the affection of noise is less relatively, the modeling result is better.

- (4) When there is noise, the various outputs having been extracted are

$$\bar{Y} = Y + A^{-1}\xi$$

where $\bar{Y} = [\bar{y}_1, \dots, \bar{y}_n]^T$, $Y = [y_1, \dots, y_n]^T$ and $\xi = [\xi_1, \dots, \xi_n]^T$ are the estimation value vector, real value vector and noise vector of various outputs, respectively. To reduce the noise affection, there is one solution that is to depress $A^{-1}\xi$, which can be achieved by two ways: one is that the absolute value of noise is small; the other is to enlarge the value of $\{k_1, k_2, \dots, k_n\}$ of A , which means enlarging the magnitude of inputs. The latter is useful when the background noise fluctuates not too much.

- (5) Since both the linear transfer function estimation and bispectrum estimation are sensitive to the accuracy of FFT, the experimental data should be processed using the segmenting, windowing and averaging technique to improve the modeling precision.

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